A Correlation-Margin Budget for the V59 Residual

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Abstract

The residual endpoint in the twin-prime analytic route contains a signed scalar and an orthogonal-complement radius. A small scalar is useful only if its correlation with the radius is quantitatively protected. We define $m = |C_\perp|/R$ and prove an exact conditional compiler: if the scalar has effective saving σ and $m \gg x^{-\eta}$, then the endpoint has effective saving $\sigma - \eta$. For the current baseline exponent $E_0 = 5/3$ and target $E_* = 1997/1200$, the strict payment is $\sigma - \eta > 1/400$. A sharp two-dimensional construction proves that a negative phase sign alone gives no positive lower bound on m. Finally, we convert the TPC-271 finite certificate to exact rational m^6 intervals: the $96 \to 192$ sixth-power margin ratio is below $(1/32)^6$ while the phase sign remains negative. The result is a conditional theorem and a finite numerical audit, not a source-level estimate or a twin-prime proof.

1 Position and scope

The TPC chain studies a literal finite V59 operator with prime shell, unit masks, deleted diagonal, source β, a registered growing cutoff, and a rank-three block projection. TPC-271 put the signed residual scalar and the two residual norm lanes into one certificate. Its exact coordinates were

$$C_\perp = \langle (I-P_3)w, (I-P_3)g\rangle, \quad W_\perp = \|(I-P_3)w\|^2, \quad G_\perp = \|(I-P_3)g\|^2,$$

$$R^2 = W_\perp G_\perp, \qquad \mathsf{Xi} = \frac{(W_\perp G_\perp)^3}{N^{10}}, \qquad \mathsf{Xi}_C = \frac{|C_\perp|^6}{N^{10}}.$$

The present paper asks which additional hypothesis would turn a signed scalar estimate into an endpoint estimate, and whether the observed phase label supplies that hypothesis.

The claim ceiling is explicit:

$$\texttt{PROVED\_CONDITIONAL}, \quad \texttt{PROVED\_EXACT}, \quad \texttt{NUMERICALLY\_CERTIFIED},$$

with source-level margin, arithmetic L^2, full Gate B, and the twin-prime conclusion all still open.

2 The correlation coordinate

On a row with positive residual lanes define

$$R = \sqrt{W_\perp G_\perp}, \qquad m = \frac{|C_\perp|}{R}.$$

By Cauchy–Schwarz, $0 \le m \le 1$; the paper works with $m > 0$. The sixth-power coordinates eliminate the algebraic square root.

Lemma 1 (exact margin identity)**.** *On every positive row with $C_\perp \neq 0$,*

$$m^6 = \frac{\mathsf{Xi}_C}{\mathsf{Xi}}, \qquad \frac{\mathsf{Xi}}{\mathsf{Xi}_C} = m^{-6}.$$

Proof. Substitution gives

$$m^6 = \frac{|C_\perp|^6}{(W_\perp G_\perp)^3} = \frac{|C_\perp|^6/N^{10}}{(W_\perp G_\perp)^3/N^{10}} = \frac{\mathsf{Xi}_C}{\mathsf{Xi}}.$$

The reciprocal identity is immediate. All divisions are by positive quantities. □

This identity gives a practical certificate: outward intervals for Xi_C and Xi yield an outward interval for m^6, with no floating-point root extraction.

3 The endpoint budget

We now state the new analytic bridge independently of the finite experiment. Let $E_0 = 5/3$, $E_* = 1997/1200$, so that

$$E_0 - E_* = \frac{1}{400}.$$

Theorem 1 (conditional correlation-margin compiler)**.** *Suppose that for all sufficiently large x, with fixed $A, b > 0$, σ, and $\eta \geq 0$,*

$$|C_\perp(x)| \leq Ax^{E_0-\sigma+\varepsilon}, \qquad m(x) \geq bx^{-\eta-\varepsilon}. \tag{2.1}$$

Then

$$|C_\perp(x)| + R(x) \leq A(1+b^{-1})x^{E_0-\sigma+\eta+2\varepsilon} \tag{2.2}$$

after an inessential enlargement of the constant. In particular, the endpoint beats E_* with a positive power saving whenever

$$\sigma - \eta > \frac{1}{400}. \tag{2.3}$$

If a raw scalar estimate has saving δ_c and loss λ_c, then $\sigma = \delta_c - \lambda_c$, and the condition is $\delta_c - \lambda_c - \eta > 1/400$.

Proof. The definition of the margin gives $R = |C_\perp|/m$. Applying (2.1),

$$R(x) \leq Ab^{-1}x^{E_0-\sigma+\eta+2\varepsilon}.$$

Adding the scalar term proves (2.2), after increasing the constant for $x \geq 1$. Since $E_0 - E_* = 1/400$, the exponent in (2.2) is strictly below E_* after choosing ε smaller than the strict gap in (2.3). No assertion is made here that either hypothesis in (2.1) holds for the growing V59 object. □

The theorem isolates the missing payment. A phase or sign statement without an exponent for m has no place in (2.3); conversely, a scalar saving alone is insufficient because the radius can be larger by m^{-1}.

4 Sharp sign-only converse

Proposition 1 (two-dimensional sharpness)**.** *For every $W, G > 0$ and every $0 < m \le 1$, there are real vectors with $\|w\|^2 = W$, $\|g\|^2 = G$, negative scalar phase, and margin exactly m. Consequently, phase sign and the two norm lanes imply no positive lower bound for the margin.*

Proof. In $\mathbb{R}^2$, take
$$w = (\sqrt{W}, 0), \qquad g = \sqrt{G}\,(-m, \sqrt{1-m^2}).$$
Then $\langle w, g\rangle = -\sqrt{WG}m$, while the squared norms are W and G. Hence $R = \sqrt{WG}$ and $|\langle w, g\rangle|/R = m$. The scalar is strictly negative for every allowed m, and $m \to 0$ makes the radius-to-scalar ratio diverge. This realizes the full interval of possible margins and proves sharpness. □

The proposition is a structural obstruction, not a counterexample to the literal V59 source. It says exactly which extra source-level statement must be supplied before the conditional compiler can be activated.

5 Finite margin audit

The producer consumes the committed TPC-271 rows without changing their operator or cut-off. It forms $\mathsf{Xi}_C/\mathsf{Xi}$ by positive rational interval division. The following table displays decimal approximations; the committed JSON stores exact rational endpoints.

row	interval for m^6 (display)	classification
64	$[4.1866, 4.1888] \times 10^{-4}$	$m > 1/8$
96	$[7.2423, 7.2455] \times 10^{-4}$	$m > 1/8$
128	$[3.7810, 3.7841] \times 10^{-6}$	$1/32 < m < 1/8$
192	$[1.4608, 1.4755] \times 10^{-13}$	$m < 1/32$
256	$[1.4959, 1.4997] \times 10^{-9}$	$1/32 < m < 1/8$
384	$[9.0122, 9.0325] \times 10^{-10}$	$m < 1/32$

Table 1: Base-row sixth-power margin intervals. The thresholds are compared exactly in the certificate.

For the four dyadic base pairs, the sixth-power margin ratios are approximately
$$0.00903, \quad 2.02 \times 10^{-10}, \quad 3.96 \times 10^{-4}, \quad 6.15 \times 10^{3}.$$
The second interval is strictly below $(1/32)^6$, while its two endpoint phase labels are both `NEGATIVE_REAL_AXIS`. The last interval is strictly above 4^6. Thus the finite data provide a particularly clean hostile control for the logical distinction between phase sign and quantitative margin.

The parent certificate also contains three $\theta = 1/2$ profile controls; the new audit retains them and records nine rows in total. The independent checker reconstructs every fraction from the parent fields, and a separate stress script rejects mutations of the collapse threshold, phase lock, conditional strictness, converse status, and fixed-power firewall.

6 Route status and limitations

The new result has three deliberately different types:

1. The margin identity and the two-dimensional converse are exact proofs.

2. The endpoint compiler is a conditional theorem whose hypotheses are still open for the source-level V59 sequence.
3. The nine-row margin table is a numerically certified finite audit.

No finite ratio supplies fixed-power credit. In particular, the observed negative phase is not an eventual phase sector, the finite margin collapse is not an asymptotic lower-bound refutation, and the compiler is not an arithmetic L^2 estimate. Route B therefore advances only in the scoped conditional sense; full Gate B and the twin-prime conclusion remain open.

7 Conclusion

TPC-272 turns the qualitative phrase "phase–radius coupling" into an explicit budget. A source-level scalar saving must be accompanied by a quantitative margin lower bound, and the two payments combine as $\sigma - \eta$, with strict cost $1/400$. The exact converse shows why the sign census from TPC-271 cannot fill this gap. The finite certificate confirms the warning: the strongest registered phase-preserving transition loses more than a factor 32 in the margin while the radius lane remains uncontrolled. The next admissible task is therefore a source-level margin lower-bound audit, not a phase-only promotion.

References

[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.

[2] L. Wang, "Phase–Radius Decoupling in a Finite V59 Residual," TPC-271 project release, 2026.